Foundations of Pelvis and Perineum

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Block: Endo/Uro/Repro

Thread: Anatomy

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Textbook

Moore's Clinically Oriented Anatomy, 9th ed; Go To: Chapter 6: Pelvis and Perineum

Learning Objectives

By exam time, you should be able to do the following:

1. Discuss the skeletal elements defining the pelvic cavity including osseous features, the differences driven by hormonally based development, and their clinical relevance
2. Discuss the joints and ligaments of the pelvic girdle and their clinical significance
3. Discuss structural and functional anatomy of muscles of the pelvic floor and the issues stemming from dysfunction
4. Discuss the fascia defining and organs contained in the perineum

Study Questions

1. A 17- year-old with a career goal of becoming a professional ballet dancer is in your office with a chief complaint of right buttock pain. The buttock pain has progressively increased over the past 2 weeks since he has been intensely training for an upcoming audition. He has been working extensively on a turn-out (actively maintaining lateral rotation of the lower limb). He also reported a deep ache in the right buttock when sitting and has noticed right lower limb medial rotation seems restricted since the buttock symptom began. What is a likely cause of these signs and symptoms?

2. A 37-year-old patient reported during her annual physical examination that she has noticed some urine leakage in the past few months. She stated that the leakage occurs when she coughs, laughs, or lifts her toddler. What is a possible reason for this leakage and what is your initial plan of care?

3. Describe the function of the perineal membrane.

A. Pelvic Boundaries

1. The pelvis is continuous with the abdominal cavity
2. There is an anatomic inlet and outlet
3. There is a greater and lesser pelvis
    a. The lesser pelvis is the true pelvis

B. Pelvic Girdle

1. The pelvic girdle bears the weight of the trunk, serves as a boney attachment for many muscles, and protects the pelvic viscera.
    a. The pelvic girdle is made of bones from the axial and appendicular skeletons.
2. The pelvic girdle includes two hip bones/ ox coxae and the intervening sacrum (1) and coccyx (1).

C. Hip Bone

1. The hipbone (os coxae) are composed of the ilium, ischium and pubic/ pubis bones.
    a. On the lateral part of the pelvis where all three parts of the os coxae fuse, there is a depression where the head of the femur sits called the acetabulum
2. Osseous features of the **ilium**:
    a. Medial view:
        i. Arcuate line
        ii. Iliac fossa
        iii. Auricular surface (sacro-iliac articulation)
        iv. Iliac crest,
        v. Posterior superior (PSIS) and inferior (PIIS) iliac spines

   b. Lateral View:
      i. Iliac crest
      ii. Ala ("wing")
      iii. Gluteal lines
      iv. Anterior superior (ASIS) and inferior (AIIS) iliac spines
3. Osseous features of the **ischium**:
   a. Ischial spine
   b. Ischial tuberosity – the ischium is known as the sits bone because it is placed most inferiorly on the os coxae where we sit. The main part we sit on is the ischial tuberosity.
   c. Ischial ramus
4. Osseous features of the **pubic/ pubis bone**:
   a. Superior pubic ramus
      i. Pectineal line
   b. Pubic symphysis
   c. Inferior pubic (ischiopubic) ramus
   d. Pubic tubercle

5. The greater pelvis is defined by the ala of the ilium and extends to the linea terminalis/Pelvic inlet
6. The lesser or true pelvis is inferior to the greater pelvis
7. The margin of the pelvic inlet is the linea terminalis
8. The outlet is below and bordered by the coccyx, pubic symphysis, ischial tuberosities and ligaments
9. The pubic arch is formed at the symphysis
   a. Estrogen generates an angle greater than 80
   b. Testosterone generates and angle less than 70

10. Other estrogen driven pelvic girdle differences include
    a. Shallower greater pelvis
    b. Thinner ilium
    c. Rounder pelvic inlet
    d. Less curvature to the sacrum
    e. Wider pelvic outlet
    f. Oval obturator foramen
    g. Small acetabulum
    h. Greater sciatic notch is larger

D. Pelvic Girdle Joints

1. Lumbosacral Joint
    a. Cartilaginous joint
    b. Articulation between vertebral body of the most superior sacral vertebra and the 5th lumbar vertebra
    c. Ligaments strengthening the joint:
        i. Anterior longitudinal
        ii. Iliolumbar
2. Sacroiliac Joint
    a. Synovial joint
    b. Articulation between the sacrum and ilium
    c. Ligaments strengthening the joint:
        i. Anterior and posterior sacroiliac
        ii. Sacrospinous and sacrotuberous

3. Pubic Symphysis
    a. Cartilaginous joint
    b. Articulation between the two pubic bones
    c. There is an **interpubic** disc within this joint
    d. Ligaments strengthening the joint:
        i. Superior pubic ligament
        ii. Inferior (arcuate) pubic ligament
4. Sacrotuberous and Sacrospinous Ligaments
    a. These ligaments with the greater and less sciatic notches create greater and lesser sciatic foramen.
        i. Greater sciatic foramen – allows communication from the pelvis to the gluteal region
        ii. Lesser sciatic foramen – allows communication from the gluteal region to the perineum
5. Obturator membrane and Canal
    a. The obturator membrane covers the obturator foramen creating a small canal superiorly within the foramen.
    b. The obturator canal allows for communication of structures from within the pelvis to the lower limb.

E. Pelvic Foramen

1. Greater sciatic foramen allows communication between pelvic cavity and gluteal region
2. Lesser sciatic foramen communication between the pelvic cavity and the perineum
3. Obturator foramen allows passage to the anterior thigh

F. Pelvic Diameters

1. Knowledge is needed to assess for vaginal deliveries
2. Transvers diameter at widest part of the inlet
3. Interspinous diameter between the ischial spines
4. True OBSTETRIC conjugate diameter is the narrowest FIXED distance a baby's head must pass through
    a. Unaffected by pregnancy hormones
5. Diagonal Conjugate is the only one measurable on physical exam, use this to estimate true conjugate

G. Pelvic Wall Muscles

1. The obturator internus originates on the obturator foramen and attaches to the femur.
    a. It is also a lateral rotator of the hip
    b. Exits the lesser sciatic foramen
    c. L5-S2
2. The **piriformis** originates on the sacrum and attaches to the femur.
    a. Exits greater sciatic foramen
    b. S1-S2
    c. Lateral rotation, abduction, stabilization of femur at acetabulum
3. The pelvic floor provides a muscular boundary between the true pelvis and the perineum. It consists of the coccygeus and levator ani.
4. The levator ani muscle group includes the puborectalis, pubococcygeus, and iliococcygeus. This muscle group hangs from a thickened portion of the obturator internus fascia called the tendinous arch.
5. The foramen are weaknesses in the pelvic wall musculature

6. Greater sciatic
    a. Formed by the Sacrospinous ligament
    b. Piriformis runs through
    c. Superior gluteal NV bundle
    d. Inferior gluteal NV bundle
    e. Sciatic nerve, nerves and vessels feeding the femoral muscles
7. Lesser sciatic
    a. Formed by the Sacrotuberous ligament
    b. Obturator internus muscle runs out
    c. Pudendal nerves and vessels traverse
8. Obturator Canal
    a. Pierces through the obturator muscle and membrane
    b. Obturator nerve and vessels going to medial thigh muscles
9. The levator ani is a muscular sling that supports the pelvic viscera and plays a major role in urinary and fecal continence.
    a. In particular, the puborectalis muscle plays an important role in fecal continence. Contraction of this muscle pulls the anorectal junction anteriorly creating a bend. Relaxation of this muscle straightens this junction and facilitates defecation.
    b. In general, weakness of the levator ani can produce leakage of feces or urine.
    c. Prolapse of internal organs can also occur
    d. Childbirth can significantly weaken the levator ani muscles
    e. Kegel exercises can reverse weakness

H. Perineum

1. The diamond shaped external surface is typically divided into a urogenital triangle and an anal triangle. A line between the two ischial tuberosities divides the diamond into two triangles.
2. The perineal body sits in the middle of the diamond and is a thickened confluence of all muscles and fascias
3. The urogenital triangle includes the external genitalia, the vagina and vulva or the root of the scrotum and penis.
4. The anal triangle includes the anus and anal canal.

I. Anal Triangle

1. Deep space on either side of the anal canal
2. Filled with fat
3. Pudendal canal runs along the lateral walls in the obturator fascia
4. Absences can occur and fistulas between fossa and anus
5. External hemorrhoids occur along the skin of the area

J. Fascia of the Urogenital Triangle

1. Superficial layer is continuous with the abdominal wall skin
    a. Scarpa fascia becomes Colles fascia in both sexes and is continuous with the thigh fascia. Scarpa also invests the scrotum as Dartos fascia
2. Deep layer is continuous with the deep fascia of the abdominal wall muscles
    a. Becomes the perineal membrane in both sexes
    b. Continuous with Buck's fascia around the penis

3. Perineal Membrane supports the urogenital triangle and attaches to the ischial rami
    a. Divides the urogenital triangle into two pouches. There are hormonal differences to what is contained in each
    b. Superficial holds the external genitalia
    c. Deep houses a portion of urethra

K. Estrogen Dependent Features

1. Surface anatomy / Vulva
    a. Mons pubis,
    b. Labia majora
    c. Labia minora
    d. Vestibule of vagina
    e. Clitoris
    f. External urethral orifice
    g. Vaginal orifice
2. Superficial pouch
    a. Crura of clitoris and covering muscle - ischiocavernosus
    b. Bulbs of vestibule and covering muscle - bulbospongiosus
    c. Superficial transverse perineal muscles
    d. Greater vestibular glands
    e. Branches of internal pudendal vessels
    f. Branches of pudendal nerves

3. Deep perineal pouch
    a. Portion of the urethra
    b. External urethral sphincter muscle
    c. Portion of the vagina
    d. Deep transverse perineal muscles
    e. Branches of pudendal nerve, internal pudendal arteries and veins
    f. Anterior recesses of the fat filled ischioanal fossae
4. The greater vestibular gland can become infected
    a. Bartholinitis
    b. Very painful and can get very large 4-5cm
5. This region can be anaesthetized during childbirth
    a. Ilioinguinal block at the round ligament
    b. Pudendal block using ischial spine as a landmark
6. Episiotomies can be performed to open up the space for childbirth
    a. Can weaken the pelvic diaphragm and perineal body

L. Testosterone Dependent Features

1. Surface anatomy Penis and Scrotum
    a. Penis glans/foreskin
    b. Body
    c. Midline raphe
    d. Scrotal sacs
    e. Scrotal raphe
2. Superficial pouch
    a. Bulb and crura of the penis and muscles
    b. Bulbospongiosus
        i. Innervation: perineal n.,
        ii. branch of pudendal nerve (S2-S4)
        iii. Expels last drops of urine/semen

   c. Ischiocavernosus
      i. Innervation: perineal n.,
      ii. a branch of pudendal nerve (S2-S4)
      iii. Compresses deep dorsal vein
      iv. Forces blood into distal penis
      v. Helps maintain erection
      vi. More developed related to penis vs. clitoris
   d. Superficial transverse perineal muscles
   e. Branches of internal pudendal vessels
   f. Branches of pudendal nerves
3. Deep Pouch
   a. Membranous portion of the urethra
   b. External urethral sphincter muscle
   c. Bulbourethral glands
   d. Deep transverse perineal muscles
   e. Branches of pudendal nerve, internal pudendal artery and vein
   f. Anterior recesses of the fat filled ischioanal fossae
4. Penis
   a. Root and bulb
   b. Body has three erectile spaces
      i. 2 corpus cavernosum
      ii. 1 corpus spongiosum
   c. Glans is an expansion of the cavernosum
5. Penile neurovasculature
   a. All are derived from the pudendal nerve and internal pudendal vessels
   b. Deep dorsal vessels and nerves run under Bucks fascia
      i. Deep dorsal veins drain erectile tissues
   c. Superficial dorsal vein under the skin and drains skin
   d. Deep arteries in the cavernosum and gives off helicine arteries
6. Scrotum houses the testicles and spermatic cord
   a. Spermatic cord coverings are extensions of the anterior abdominal wall muscles that are extended as the gonads descend

7. Trauma to the perineum may damage the urethra.
    a. Damage above perineal membrane has blood and urine collecting in pelvic cavity and /or deep perineal pouch
    b. Damage in the penile portion may have blood and urine collecting in scrotum and can extravasate up the anterior abdominal wall under Scarpa fascia

Answers to Study Questions

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

1. A 17- year-old with a career goal of becoming a professional ballet dancer is in your office with a chief complaint of right buttock pain. The buttock pain has progressively increased over the past 2 weeks since he has been intensely training for an upcoming audition. He has been working extensively on a turn-out (actively maintaining lateral rotation of the lower limb). He also reported a deep ache in the right buttock when sitting and has noticed right lower limb medial rotation seems restricted since the buttock symptom began. What is a likely cause of these signs and symptoms?

   Answer: The patient very likely has piriformis syndrome. The piriformis is a deep muscle in the buttock that assists with lateral rotation of the hip (turn-out). Overworking this muscle can cause muscle spasm. The patient may experience an ache sensation due to the spasm. The increased tone present with muscle spasm can also restrict passive elongation of the muscle (in this case medial rotation). Piriformis syndrome is typically treated by modifying the patient's activities (reducing active lateral rotation of the hip), passive stretching (into medial hip rotation), and deep tissue massage. Steroid injection is sometimes performed in calcitrant cases.

2. A 37-year-old patient reported during her annual physical examination that she has noticed some urine leakage in the past few months. She stated that the leakage occurs when she coughs, laughs, or lifts her toddler. What is a possible reason for this leakage and what is your initial plan of care?

   Answer: One possible cause of urinary leakage when coughing, laughing, or lifting is stress incontinence. Weakening of the pelvic floor can occur during or after a pregnancy or simply as part of the aging process. Stress incontinence can also occur in men. Some patients will have great success by performing pelvic floor exercises to strengthen the region. Patients sometimes find it difficult performing a pelvic floor contraction. To help them understand the movement they can be instructed to contract their muscles so the anus is elevated. Another verbal description of the movement is to contract your pelvic floor as if you were trying to stop the flow of urine while you are voiding.

3. Describe the function of the perineal membrane.

Answer: The perineal membrane is a fascial sheet that covers the anterior region of the pelvic outlet and closes the urogenital triangle of the perineum. It is attached to the ischiopubic rami. It serves as the floor of the deep perineal pouch and the roof of the superficial perineal pouch. The urethra passes through this membrane in everyone and in estrogen-dependent individuals the vagina also perforates this structure.